

Bias Mitigation in ACSEmployment Prediction

CS340 Final Project

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Task and Evaluation Setup

Prediction Task

- Dataset: ACSEmployment, CA and TX, 2018
- Target: binary employment status
- Sensitive attribute: sex
- Held-out evaluation set: 129,383 individuals

Metric Scope

- 13 TFMA metrics
- 5 extended fairness metrics
- Focus: group-level reliability and error gaps

Goal: reduce gender-related fairness gaps without losing too much predictive utility.

Baseline Metrics by Sex

Metric	Female	Male	Overall
Sample Size	65,819	63,564	129,383
AUC	0.8713	0.9244	0.9016
Binary Accuracy	0.7889	0.8527	0.8202
Positive Rate	0.4668	0.5253	0.4956
Negative Rate	0.5332	0.4747	0.5044
Precision	0.7216	0.8267	0.7763
Recall (TPR)	0.8059	0.8853	0.8481
True Negative Rate	0.7767	0.8213	0.7971
False Positive Rate	0.2233	0.1787	0.2029
False Negative Rate	0.1941	0.1147	0.1519
False Discovery Rate	0.2784	0.1733	0.2237
False Omission Rate	0.1521	0.1186	0.1366

Base model performance metrics by sex

Fairness Metric	Value/Difference	Threshold	Status
Disparate Impact (80% Rule)			
Female Positive Rate	0.4668		
Male Positive Rate	0.5253		
Ratio	0.8886	≥ 0.80	Fair
Average Odds Difference	0.0620	≤ 0.10	Fair
Predictive Parity			
Female Precision	0.7216		
Male Precision	0.8267		
Difference	0.1051	≤ 0.10	Unfair
NPV Equality (Cond. Use Accuracy)			
Female NPV	0.8479		
Male NPV	0.8814		
Difference	0.0335	≤ 0.10	Fair
Performance Gaps (AUC)			
Female AUC	0.8713		
Male AUC	0.9244		
AUC Gap	0.0531	≤ 0.05	Unfair

Base model extended fairness metrics summary

Baseline Bias Pattern

What the Baseline Gets Right

- Disparate Impact is within the 80% rule: 0.8886
- Average Odds Difference is acceptable: 0.0620
- NPV Equality is also within threshold: 0.0335

Main Takeaway

The baseline is not uniformly unfair; the largest issues are concentrated in prediction reliability and ranking.

Where the Gaps Remain

- Predictive Parity gap: 0.1051 (> 0.10 threshold)
- AUC gap: 0.0531 (> 0.05 threshold)
- Female precision is lower than male: 0.7216 vs. 0.8267

Error Pattern

Female FPR and FNR are both higher than male: 0.2233 vs. 0.1787 and 0.1941 vs. 0.1147.

Mitigation should target error-rate parity and representation bias, not just positive rate balance.

Mitigation Method

1. Adversarial Debiasing

Gradient reversal discourages gender information in hidden representations.

2. Equalized Odds Loss

Directly penalizes FNR and FPR differences across groups.

3. Sample Re-weighting

Balances gender groups and boosts positive examples by 1.5x.

$$\mathcal{L}_{main} = \mathcal{L}_{BCE}(y, \hat{y}) + \lambda_{fair} (|FNR_f - FNR_m| + |FPR_f - FPR_m|)$$

$$\lambda_{adv} = 0.5, \quad \lambda_{fair} = 0.5, \quad \text{Adam optimizer, 25 epochs}$$

Double-Win Scorecard

Metric	Base F	Fair F	Base M	Fair M	Result
<i>Auto-pass (teacher policy)</i>					
Example Count	65,819	66,809	63,564	62,574	WIN
<i>Double wins – both groups improve</i>					
False Negative Rate	0.1887	0.0814	0.1139	0.0666	WIN
False Omission Rate	0.1488	0.0816	0.1180	0.0830	WIN
Negative Rate	0.5302	0.4165	0.4735	0.3936	WIN
Positive Rate	0.4698	0.5835	0.5265	0.6064	WIN
Recall (TPR)	0.8113	0.9186	0.8861	0.9334	WIN
True Positive Rate	0.8113	0.9186	0.8861	0.9334	WIN
<i>Double losses – both groups regress</i>					
AUC	0.8716	0.8615	0.9242	0.8994	LOSS
Binary Accuracy	0.7903	0.7665	0.8523	0.8188	LOSS
False Discovery Rate	0.2784	0.3420	0.1744	0.2449	LOSS
False Positive Rate	0.2247	0.3428	0.1803	0.2915	LOSS
Precision	0.7216	0.6580	0.8256	0.7551	LOSS
True Negative Rate	0.7753	0.6572	0.8197	0.7085	LOSS
Total Double Wins				7 / 13	

Extended Fairness Metrics Improved

Metric	Base Model	Fair Model	Direction
Disparate Impact Ratio	0.8886	0.9622	closer to 1
Average Odds Difference	0.0620	0.0330	lower
Predictive Parity Gap	0.1051	0.0971	lower
NPV Equality Gap	0.0335	0.0014	lower
AUC Gap	0.0531	0.0378	lower

All five extended fairness metrics move in the desired direction.

Overall Experimental Result

Improved Metrics

- 12 metrics improved in total
- 5 extended fairness metrics improved
- 7 TFMA metrics improved

The improvement is targeted: fairness gaps and missed-positive errors improve, but predictive ranking and precision costs remain measurable.

Performance Trade-off

Metric	Base	Fair	Change
AUC	0.9016	0.8784	-0.0232
Accuracy	0.8202	0.7922	-0.0280
Precision	0.7763	0.7066	-0.0697
Recall / TPR	0.8481	0.9265	+0.0784
FNR	0.1519	0.0735	-0.0784
FPR	0.2029	0.3193	+0.1164

Interpretation

- The model predicts more positives after mitigation.
- Fewer positive cases are missed.
- More false positives are introduced.

Main Trade-off

Fairness and recall improve, while accuracy, AUC, and precision decrease.

Conclusion

- Adversarial debiasing, Equalized Odds loss, and sample re-weighting improve all five extended fairness metrics.
- The fair model reduces the AUC gap from 0.0531 to 0.0378 and predictive parity gap from 0.1051 to 0.0971.
- Recall improves from 0.8481 to 0.9265, reducing missed positive cases.
- Accuracy, AUC, precision, and FPR show that fairness gains come with real performance trade-offs.

Questions?